AI ETHICS

AI Ethics: The Case for Including Animals

By Peter Singer, Ira W. DeCamp Professor of Bioethics, University Center for Human Values, Princeton University and Yip Fai Tse, Research Assistant, University Center for Human Values, Princeton University

In this module, Singer and Tse examine the existing AI ethics narrative and find a distinct lack of consideration for animals in the current discourse. Drawing on numerous case studies from the field, they make the case for including animals in the AI ethics debate. They also review existing applications of AI on animals and identify the ways in which practitioners can incorporate considerations for animals in the development and implementation of ethical AI technologies.

Lecture Transcript

Singer 0:01 Hello, I'm Peter Singer. I'm a professor of bioethics at Princeton University, and the author of *Animal Liberation*, as well as *Practical Ethics*, and other books. I'm talking to you about AI ethics and the case for including animals, together with Tse Fai.

Tse 0:21 Hello, I'm Fai. I'm based in Hong Kong. I'm a research assistant to Peter Singer at Princeton University.

Singer 0:30 Here's an outline of what we're going to be talking about in this talk. You can see, we're going to start by saying that there is something missing in the ethics of artificial intelligence, as it is discussed today. We're going to argue that it should include animals. We're going to talk about how AI impacts on animals and give you some case studies of those ethical implications. And finally, we'll ask what can you do about the situation that we've described?

Singer 1:10 So let's begin with the idea that there is something missing in AI ethics, as it is standardly done today. You can probably already guess what that might be. There are many issues discussed in AI ethics, such as AI safety, algorithmic bias, problems about privacy and transparency. What impacts will AI have on employment? Will it foster greater inequality? What about military use of AI? And of course, many people are concerned about the threats to the very survival of humanity, of our species, arising from the possibility of artificial general intelligence, and more particularly, super intelligent AGI. And more recently, we've had quite a bit of discussion about the moral status of possibly conscious or sentient artificial intelligence. But one category of sentient beings is virtually never mentioned. And that category is, of course, animals.

Singer 2:22 We did some research on this. We looked at 71 courses in AI ethics or computer ethics. And in none of those, was there concern about AI's impact on animals as individuals. There was some concern about AI's impact on endangered species, causing extinction of species, and in general on sustainability issues, but nothing on what we might think of as animal welfare or animal rights, questions about individual animals. We also looked at more than 300 papers and books on AI. And we found only 4 of those 300 plus papers and books actually referred to the impact of AI on animals. We also looked at 68 statements of principles or statements of AI ethics that have been issued by various bodies and individuals. Only 2 of them mentioned AI's impact on animals. They mentioned sentient beings and AI's possibility of AI being sentient beings, but not specifically animals. And in fact, the welfare of one at this stage, purely hypothetical, conscious machines gets mentioned far more often than the actual situation where AI is having an impact on non-human sentient beings.

Singer 4:10 Let's ask then why ethics should include animals. I go back here to Jeremy Bentham, the founding father of the Utilitarian tradition. Bentham said in the late 18th century, the question is not can they reason nor can they talk, but can they suffer? We agree with Bentham. We think that the crucial question as to whether a being should be included in our ethical considerations, if you like whether a being has moral status or moral standing, is not whether that being can reason or talk, but whether that being can suffer. And of course, it therefore follows that if machines can suffer they matter too.

Singer 5:05 But we do know that animals can suffer. But many non-human animals anyway can experience pain and pleasure. And we regard the refusal to take that pain and pleasure into account as speciesism. That's a term that we use to make the parallel between racism and sexism, which we're familiar with, and which I'm sure that those of you who are listening to this already dissociated with and would condemn racism and sexism. But speciesism, too, is a prejudice against taking seriously the interests of beings who, if you like, are not us, who we regard as the other. And because we are more powerful than they are, in just the same way that for example, Europeans were more powerful than Africans when they enslaved them. And in just the same way, as men have long been more powerful than women. So just as those dominant groups exploited and used, the others, the non whites or the non males, for their own purposes. So we believe that today, we are exploiting and using non-human animals for our own purposes in a way that is not ethically defensible. Instead, we argue that we should be giving equal consideration to the interests of animals, that is where they have similar interests to our own. The fact that they're animals should not lead us to discounting or giving less weight to their interests. And that's irrespective of the species of the being whose interests we are concerned with.

Singer 7:04 We approach this topic from a broadly utilitarian viewpoint as Jeremy Bentham did. But it's not necessary to be a utilitarian to take this view, and there are many proponents of non utilitarian ethical views, who also think that animals matter. For example, among deontologists, Christine Korsgaard is perhaps the leading Kantian ethicist in the English language at present. And she thinks that Kant made a mistake in not including animals, and her version of Kantian Ethics does consider that animals matter and we have to give way to their interest. Tom Regan is perhaps the best example of someone who takes a rights-based point of view for animals. Regards animals as having rights and therefore as having moral status. Mark Rowland is a Contractualist, you could say he's working in tradition that goes back to Hobbes and Locke and Rousseau, but also, in more recent times, to the Harvard philosopher John Rawls, but he thinks that contractualism also has to include animals and give them moral status. Among those who take a feminist ethical perspective, particularly the form of feminism, known as care ethics, Josephine Donovan has insisted that animals must have a place in a ethics of care. And of course, in non Western philosophies we have had for millennia, ethical views, such as Buddhism and Jainism, that also give a face to animals and say, we should have compassion for all sentient beings.

Singer 9:03 Let's ask then, how AI does impact animals? Because of course, if AI didn't impact animals, it would be reasonable not to discuss that impact ethically. But in fact, it does impact animals, and often it impacts animals in a negative manner. For example, AI or robots using AI may increase the efficiency of factory farms, and we will discuss that in more detail shortly. Some drones are deliberately designed in order to kill animals. Self driving cars may hit animals and not seek to avoid them unless hitting them would pose a risk to passengers or to the car. And AI systems may actually propagate speciesism in a variety of actions that affect animals.

Singer 10:01 AI can also impact animals positively. For example, self-driving cars could be programmed to avoid animals. AI could design ways of replacing the use of live animals in experiments, often very painful use. AI can be used to better understand animals, in a sense to tell us what they want or need. And AI can assist research in alternative proteins which may replace factory farming.

Singer 10:37 Let's look at some of the ways in which AI interacts with animals. We can distinguish designed or intended interactions from unintended interactions. And among design interactions, we can have robots designed to care for pets, we can have AI managing zoos. We've mentioned drones designed to kill animals, and AI in factory farming is obviously designed to interact with animals. Among unintended interactions, self-driving cars may hit or may avoid animals. And domestic robots, just looking after a house for example a home, may have interactions with animals who come into that home unexpectedly, such as rats or wild animals that get into the home. There are also AI impacts where there is no direct interaction, but there are indirect impacts. For example, AI may reinforce speciesist language. When we talk about people being “pigheaded”, we give the impression that pigs are stubborn or stupid. When we talk about people “chickening out” of some dangerous situation, we present chickens as being very timid. But also when we ask AI for something about chicken, it's likely to give us, the search engine, is likely to give us not the animal, not the live animal of a chicken, but rather the corpse of a chicken or the flesh of a chicken, giving the idea that these animals are somehow essentially or intended to be meat rather than animals with lives of their own to live. Some language models may tell us that killing some animals is okay. They may recommend actions that affect animals such as eating animal products, and they may even allow some forms of cruelty to be spread or advocated. For example, some social media platforms have videos showing animal cruelty.

Singer 13:02 Let's look now at some case studies, so a bit more detail about the things that we've just been talking about. These case studies are relevant to the real world. They show that the interaction of AI with animals is complicated and subtle. And they show that we can do better.

Singer 13:24 I'll start with speciesism in AI. If you go to Ask Delphi, Delphi gives you answers to whether certain actions are right or wrong. And here's some of the answers that we got when we went to Delphi. We asked about whether it's right or wrong to kill a cat or a dog. And we got that as wrong even if killing a dog is culturally accepted. On the other hand, killing chickens, cows, pigs, even a pig after it's lived a miserable life on a factory farm, or a pig in a slaughterhouse, all of that is considered okay. So we have a contrast here between killing animals who are really rather similar in terms of the cognitive abilities and their emotional life, some being right and some being wrong, based obviously, on the animals that we eat as distinct from the animals we have as companions. Killing a pig on the street was seen as cruel, and rather curiously, killing a fish was seen as bad, although it's hard to see why anyone would think that killing a fish is bad, but killing a cow or a pig or a chicken is not bad.

Singer 14:49 We also looked at AI trained on human texts. For example, Word2Vec is trained on Google News, Wikipedia and Twitter. And it shows the species bias when you test for the similarities of words describing humans, non-farmed animals, and farmed animals with positive adjectives. So as you see here, the pattern is the same whether we're talking about Google News, Wikipedia or Twitter. We find that there is closer proximity with farmed animals to negatives and more distance with positives. Non-farmed animals have more positive references, and humans have more still. And that doesn't say that pattern is repeated.

Singer 15:45 Now let's look at the case study of the use of killer drones designed actually to kill animals. Typically, this is done where animals are considered pests. And what we're looking at here is the brushtail possum, which is a native animal of Australia and in Australia is a protected animal. But brushtail possums were introduced to New Zealand, in order to start up a fur industry. The idea was that they would be reared and killed for their fur. But some of them escaped. They're quite smart and agile animals. And they got into New Zealand forests, which have not evolved with mammals that eat leaves like the Australian possums. And there they were causing damage to the New Zealand forest. So the New Zealand government decided to try to kill them, and is using killer drones in order to do that.

Singer 17:03 Now, is this something that is justifiable? We argue that we certainly need to take account of the deaths and suffering of the possums in order to consider that issue. It might sometimes be justified, because it might be that the suffering that the so-called pests cause is greater than the suffering caused by killing them. They might destroy the habitat for native New Zealand animals, for instance, and there might be a lot of suffering when they starve because of that loss of habitat. One might also argue that beyond just suffering, the preservation of a natural ecological system is an important value that may justify causing some suffering. But there's a question as to how much suffering that can justify. Among the philosophical questions, this raises, therefore, criteria about whether what is an invasive animal? What is a so-called pest animal? Is there a point at which we regard animals as having become native because they've been there for a long time? Whose perspective are we judging this by? And most importantly, is the welfare of the individual animals actually weighed into the decision that all things considered, it's justifiable to kill them?

Singer 18:31 In terms of killing animals, whether for conservation or other purposes, there is also the question of whether we take this too lightly when we use autonomous drones to kill. And this question has been discussed in terms of the killing of humans by drones, of course. And the ecologist Craig Morley has compared this to, sorry, has compared the killing of animals to the use of autonomous drones in killing in war. As he said, "You used to be able to see your opponent. Now you just press a button, you fire a missile, you become a little detached from the reality that you have killed something or somebody over there."

Singer 19:22 I turn now to autonomous vehicles, self-driving vehicles. All cars may hit animals. And in fact, the number of cars that do hit and kill animals is huge. According to one 2015 study, just for the United States, in one year, something between 89 and 340 million birds are killed. And that's only birds, there may well be other animals killed. And as you will see in the next slide, which one is perhaps a little unpleasant to look at, in some cases, a lot of other animals may be killed. This is a situation on a Pacific Island where crabs annually migrate across roads. And as you can see here, if cars continue to drive on those roads, they kill very large numbers of crabs, and some of them won't be killed, of course, instantly, they will be partially crushed, and will die slow and painful deaths.

Singer 20:34 Who bears the moral responsibility when an autonomous vehicle hits an animal? That may depend on just how autonomous the vehicle is. So let's focus on level five self-driving cars, which drive without any human intervention. It's ethically problematic to hold autonomous vehicles to a higher standard than humans. But for humans, it may be impossible for them actually to avoid animals. In some cases, an animal runs or a bird flies across the path of the vehicle and the human reactions are just not quick enough to avoid killing or injuring the animal. But self-driving cars have the possibility of having faster reactions and therefore better possibilities of avoiding killing animals.

Singer 21:39 Some other questions raised by autonomous vehicles. If we do design them to avoid killing animals, to be safer for animals, that may mean that there's a higher cost in training the AI to avoid animals, especially if we wanted to avoid small animals, more costly detection and computations. But we have to ask whether that's a reason to ignore our obligations towards animals? And what about the situation in which the owner of the car prefers that the car ignore animals? Or possibly even prefers that the car hit animals? Should the car system still insist on not ignoring animals and definitely not deliberately hitting animals? Is that how we ought to be designing autonomous vehicles? Here's the best solution, of course, to close the roads when there are likely to be a lot of animals on the road. So as during this red crab migration that we saw earlier, if the road is closed, then the crabs are safe. What about if in fact, it's not a crustacean but a vertebrate? Does that make any difference? Here we have a frog migration. This is taking place in China. And as you can see, if you look closely at the photo, frogs have already been crushed on the roads, and it seems likely that many more are going to be crushed. Does the fact that these are vertebrates make it worse? There is evidence that crabs and some other crustaceans can feel pain, so perhaps it's the same for both of them. I am now going to hand over to Tse Fai who is going to talk about factory farming and AI.

Tse 23:49 Thank you, Peter. Before we talk about AI's impact on factory farmed animals, we need to talk a little bit about factory farming itself. So factory farming is an industry that keeps huge numbers of animals. Just to give some statistics: there are 72 billion farmed chickens raised per year, and in terms of laying hens, there are 75 billion existing at each point in time roughly. There are 2.5 billion ruminants plus farmed pigs per year. And the numbers are even larger for animals underwater, such as farmed fish, which is at roughly 92 billion per year. And the statistics are not really very available for invertebrates such as shrimps, crabs, lobsters, octopuses, and insects, which Peter mentioned who might be sentient. And in fact the UK Government for example, recently accepted that crustaceans should be considered to be possibly sentient and not treated in unnecessarily cruel ways.

Tse 25:14 So AI is, in fact, being used in factory farming already now. It's not yet extremely popular, but it is getting experimented or actually commercially implemented. AI is used in many different ways, such as using it in identifying the animals, similar to the ways we use facial recognition to identify humans. So for animals who have faces, or that the computer scientists would like to imagine them as having faces, they can recognize the pigs' or the cows' faces or body features. They can use AI to model the growth rate to track them and then to accurately predict how they correlate with physical parameters. They can use AI to detect diseases or even further, they can even try to predict or model whether the animal will get sick in the next day or even in the next few hours using some physical parameters, or body metrics. They can use AI to optimize the feed, both the formula of the feed, and also maybe the feeding pattern in some operations, such as fish farming. They're starting to look into using AI for genetic selection. They're using AI in combination with robotics, such that they can directly handle the animals instead of just telling them, the humans, what are the solutions or the recommendations about handling animals.

Tse 27:00 Some really big players are really trying to invest in AI in animal agriculture or factory farming. You can see big names such as IBM, LG from Korea, Alibaba from China and Microsoft. These are really big players, very interested in this industry. And with AI being used in factory farms, we need to ask the question whether this is good or bad for the animals and to look into the question, we'll probably have to start with the question: What is the factory farm industry optimizing for? So as you can see from this figure, the previous use of prominent or emerging technology for factory farming was genetic selection. And the result of it was much faster growing chicken over the years. They've selected the faster growers. And you can see and there was, this was already the old figure. Chickens have over the years drastically grown faster, so fast that they grow into ways that they cannot support by the legs, and they suffer from leg problems. And that's a huge amount of suffering for the animals because of this technology. And we suspect these things could happen again for AI, and we need to be careful, and we're going to analyze it.

Tse 28:37 But a little bit more on the problems of factory farming. It is an industry that is often screened off from the public about its internal operations. Very few resources are spent on the problem to try to alleviate the suffering of the animals. That means very few charities are there to help them. And it’s not a popular topic in the media. It's not a major concern for politicians and governments also. So it's often not in the policy making field. Factory farming is problematic ethically, in our opinion. And we are best to think about it in terms of the principle of equal consideration of interests. And if we take the interest or the suffering of the animals and humans equally, we have to ask ourselves how much benefit it is to humans that is it really justified enough to eat animal products raised from factory farms for our own interests and let the animals suffer huge suffering. So is it the taste? Is it the health? Or is it keeping jobs for human? In terms of taste, probably, there is some pleasure to some humans in enjoying the meals with meat or animal products. But it seems to us that the suffering of the animals by far outweighs these pleasures. And for some vegetarians or vegans who have enjoyed vegetarian and vegan diets, they seem to enjoy a vast diversity of tasty vegan or vegetarian food. And because they stay away from meat, they are trying very deliberately to explore different flavors such as herbs or different crops.

Tse 30:54 And in terms of health, it is at least as healthy, the vegan or vegetarian diet, as some evidence suggests, seems to be at least as healthy or in some studies, more healthy than diet with meat or animal products if we do remember to supplement with vitamin B 12, which is basically the only nutrient we cannot get from plants consistently. Some people might argue, factory farming should be around to keep jobs for the humans. But it might be a type of job that is lowly paid, and in terrible conditions. Humans often suffer from trauma by seeing animals in very bad conditions, or animals can sometimes be hard to manage, and the humans might be frustrated inside. We can also ask whether the jobs can be, whether people who take these jobs can be replaced with other jobs. And presumably, if there are less people eating meat, there will be other industries such as alternative protein, which require also some plants to be crop. And some humans might be moved to some other new jobs. So we have to consider these issues.

Tse 32:32 And how much harm is there to humans may be is another important question to ask. There might be harms to humans in some more subtle ways. In eating meat, we can be having or propagating a cruel culture by continuing to eat meat. Or we might have public health issues by keeping factory farming around which has a potential to spread epidemics or even pandemics as according to some studies. And what about factory farming's impact on climate change? Because animal products, most end products have much higher carbon footprint than what you usually see in most plants crops.

Tse 33:21 And what about the energy and grain wastage by feeding a lot of feed just to get some protein from animals. For example, for beef, or pork to get some one kilogram of pork, you need a few kilograms of crops to get the protein. And for beef, the protein conversion ratio is roughly from 10, up to sometimes 20. And that is, in our perspective, similar to directly wasting the grain. And also, an even more important question is how much harm we're causing to the farmed animals, and I invite you again to think about the numbers of the animals affected and the intensity of suffering caused by factory farming.

Tse 34:23 So in summary, our view is that factory farming is morally indefensible. And that leads us to the discussion whether AI is going to be better for the farm animals? Is it going to make the situation better or worse? So there indeed might be short term benefits for the animals, as some of you might already be imagining. There will be early disease detection or detection of injuries, and that means better health for the animals. And not just that, because AI detects disease and injury earlier than humans, who might just be checking the barn around a few times a day, that means the animals will suffer less time before they identify with the diseases or injuries, and that means they will be treated or in maybe a lot of cases killed earlier, which reduced the duration of suffering. And AI cannot be frustrated or they cannot become sadistic, so they will not intentionally mistreat animals in bad ways. Granted, they will probably make some mistakes, but even the mistakes raised should be expected to be lower than humans. But we also worry that the use of AI may be worse in the long term for the animals. And some reasons to think this might be the case is that better disease control means there is a potential for the factory farmers to increase the density of the animals with the same mortality rate. And there's a huge incentive to do so because the larger the area, the more labor and the larger land cost there will be.

Tse 36:22 There are already some companies who are claiming that the AI will actually help increase the stocking density of the factory farms. That seems to be a good thing to advertise for the AI companies to increase their attraction to factory farms. There might be potential welfare problems when it comes to AI being used in genetic selection. As you can see, in a few slides before, the previous use of genetic selection using just human intelligence and human processes already came up with the fast growing chickens who suffer greatly, and look like they are very in an abnormal situation. And with AI, because AI can theoretically automate the process of genetic selection and presumably make it even more efficient and they might identify more patterns than humans, there's a worry that they might create more beings who suffer horribly, like the fast growing chickens. With higher density allowed by AI, possibly AI saving some feed costs, because they know how to better use the feed or they know the appetite of the animals so that the feeding is more accurate, so less feed will be wasted that means lower costs for them. And that means a lower price for the animal products. And according to simple economics, that should mean more animals demanded, sold on the market, and the end result may likely be more animals farmed. A lower price for animal products will also mean by simple economics that alternative proteins such as cultivating meat, or in plant based protein, they might become less cost competitive, so that the development might become slower, or the probability that they will eventually replace this horrible industry of factory farming might be lower.

Tse 38:53 So in summary, it seems to us that on some factory farms, AI will likely increase overall suffering even though there will be some short term gains for the animals. But it might not be worth it if we do not get to replace factory farming industry as soon as possible if we introduce AI to factory farming. But we can actually instead use AI to fight factory farming. AI is now being used in research in plant based meat, egg or dairy alternatives and also used in the research of cultivated meat. It is the case that AI used in these good research directions is much less invested than the AI used in factory farming. The reason is clearly that AI and factory farming look more profitable than using AI in alternative protein for the computer sciences, so they are not yet very interested. But we argue that this should be changed.

Tse 40:14 So in the final section, we would like to summarize what you can do. So, you indeed, can do something to improve the lives of animals or at least reduce harm to them with your AI systems. As someone who works on AI, you can learn about animal ethics, or ethics generally, to know more about how your AI systems will impact the world in an ethically relevant way, and how you should think about it or react to it. You should investigate your impact on animals just like you should investigate your impact on humans with your AI systems. You should try to reduce the harm you do to deal with your technology or products. You should choose projects with positive impacts and stay away from negative ones, whether we are talking about impacts on animals or impact on humans.

Tse 41:23 So to conclude, animals matter ethically. AI impacts animals. AI's impact on animals, we argue, should be identified and ethically evaluated. Animals are largely ignored in ethics now, and that is why we should start taking attention to it. And finally, we can do something to change the situation. And with that, we conclude the talk.

Singer 41:56 Thanks very much, Fai. And thank you for watching and listening to us. We do want to acknowledge thanks for financial support from the Center for Information Technology Policy at Princeton University, and also in the University Center for Human Values at Princeton. They've jointly provided a grant to support Tse Fai's research assistance on this topic, without which we would not have been able to produce this video. Thank you very much.

Tse 42:31 Thank You.